Anjali Desai

Kurnool, Andhra Pradesh, India | anjali_desai@srmap.edu.in | +91 9336473232 | LinkedIn | GitHub | Portfolio

Professional Summary

Data & Insights–oriented engineer (B.Tech CSE, AI/ML; CGPA 8.76; graduating May 2027) with SQL-first, full-stack foundations: PostgreSQL/MongoDB data layers, Python (FastAPI) pipelines, and React dashboards with real-time alerts. Built a real-time transaction scoring pipeline (**12.7 ms P95 at 111 events/sec**) and a React/TypeScript analytics app with Recharts visualizations and CSV/PDF ingestion. Ships reproducibly with Docker, Git and GitHub Actions. MongoDB and Oracle Java SE 17 certified; Winner, Microsoft Student Community Hackathon.

Technical Skills

Languages: SQL, Python, JavaScript, TypeScript, Java (SE 17), C++
Frontend & Visualization: React, Next.js, Recharts, TanStack Query, Radix UI/shadcn
Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, Socket.IO/WebSockets, Redis Streams, Apache Kafka
Databases & Data Modeling: PostgreSQL, MySQL, MongoDB, Drizzle ORM, schema design (database-per-service), CSV/PDF ingestion (PapaParse, pdf-parse)
Infrastructure & Delivery: Docker, Kubernetes, Helm, GitHub Actions (CI/CD), Git/GitHub, Linux, OpenTelemetry
Practices: Real-time alerting, root-cause troubleshooting, unit testing, technical documentation (SRS), cross-functional collaboration

Projects

RealTimeGuard – Fraud Detection Engine | *Python, FastAPI, Redis Streams, LightGBM, MongoDB, Socket.IO* GitHub

- Built an event-driven pipeline that streams and scores financial transactions and pushes **live fraud alerts** to the UI over Socket.IO, with MongoDB persistence.
- Optimized event processing and inference to reach **12.7 ms P95 latency at 111 events/sec**.
- Traced bugs end to end (UI state → API → database) to find root causes for cross-role integration issues.

QuantumShield – Full-Stack Data Web Application | *React, TypeScript, Node.js/Express, PostgreSQL, Drizzle ORM* GitHub

- Built CSV/PDF data-ingestion pipelines (Multer, PapaParse, pdf-parse) feeding a PostgreSQL schema, with Zod validation of incoming data.
- Delivered interactive **Recharts visualizations** and real-time WebSocket updates in a React front end, using TanStack Query for data fetching.

Event-Driven Order and Payment System | *NestJS, Kafka, PostgreSQL, Docker, Kubernetes, Helm* GitHub

- Designed a service-per-domain platform with a dedicated PostgreSQL schema per service and Kafka messaging.
- Added OpenTelemetry/Jaeger tracing, and automated test and deploy pipelines with GitHub Actions and Docker/Helm for fast production releases.

SmartWealth – Investment Dashboard | *Python, Streamlit, Plotly, SQLite* GitHub

- Built an interactive dashboard with **Plotly** charts of live market price trends from the yfinance API, turning financial data into visuals for non-technical users.
- Designed a **SQLite** persistence layer storing user profiles and advisory history, so users can review past reports.
- Integrated TextBlob sentiment analysis of news headlines into the Python pipeline to label stocks Bullish or Bearish.

Work Experience

Research Intern Jun 2025 – Aug 2025
SRM University, AP *Remote*

- Built a Neural Architecture Search prototype in C++/Python, running **100+ generations** of experiments across conv, pooling, dense and dropout configurations.
- Cut setup time from 4 hours to 2 hours (**2x faster** iterations) and reduced manual tuning by **30%** with a reproducible, modular search framework.
- Compared **5+ crossover/mutation strategies** and reported which found top architectures within 3 generations.

Certifications & Achievements

- **Oracle Certified Professional: Java SE 17 Developer** (Feb 2026) | **MongoDB Associate Developer, Node.js** (Apr 2026)
- **Winner**, Microsoft Student Community Hackathon: ML-driven predictive tracker using time-series data
- **Top 100 (PAN India)**, Adobe India Hackathon

Education

SRM University AP Aug 2023 – May 2027
B.Tech in Computer Science and Engineering (CGPA: 8.76 / 10.0) *Andhra Pradesh, India*